

CHAPTER 7

Structural Organisation in Animals

Epithelial & Connective Tissue

- Which of the following statements is **correct** about the type of junction and their role in our body? (2024 Re)
 - Adhering junctions facilitate the cells to communicate with each other.
 - Tight junctions help to stop substances from leaking across a tissue.
 - Tight junctions help to perform cementing to keep neighbouring cells together.
 - Gap junctions help to create gap between the cells and tissues.

- In which of the following connective tissues, the cells secrete fibres of collagen or elastin? (2024 Re)

A. Cartilage B. Bone C. Adipose tissue
D. Blood E. Areolar tissue

Choose the most appropriate answer from the options given below:

- B, C, D and E only
- A, B, C and E only
- B, C and D only
- A, C and D only

- Match **List-I** with **List-II**: (2024 Re)

List-I		List-II	
(A)	Squamous Epithelium	(I)	Goblet cells of alimentary canal
(B)	Ciliated Epithelium	(II)	Inner lining of pancreatic ducts
(C)	Glandular Epithelium	(III)	Walls of blood vessels
(D)	Compound Epithelium	(IV)	Inner surface of Fallopian tubes

Choose the **correct** answer from the options given below:

- A-II, B-III, C-I, D-IV
- A-II, B-IV, C-III, D-I
- A-III, B-I, C-II, D-IV
- A-III, B-IV, C-I, D-II

- Match List-I with List-II: (2024)

List-I		List-II	
(A)	Unicellular glandular epithelium	(I)	Salivary glands
(B)	Compound epithelium	(II)	Pancreas
(C)	Multicellular glandular epithelium	(III)	Goblet cells of alimentary canal
(D)	Endocrine glandular epithelium	(IV)	Moist surface of buccal cavity

Choose the correct answer from the options given below:

- A-III, B-IV, C-I, D-II
- A-II, B-I, C-IV, D-III
- A-II, B-I, C-III, D-IV
- A-IV, B-III, C-I, D-II

- Given below are two statements: (2023)

Statement-I: Ligaments are dense irregular tissue

Statement-II: Cartilage is dense regular tissue.

In the light of above statements choose the correct answer from the option given below:

- Statement-I is false but Statement-II is true.
- Both Statement-I and II are true
- Both Statement-I and II are false
- Statement-I is true but Statement-II is false

- Match List-I with List-II. (2023)

List-I		List-II	
(A)	Mast cells	(I)	Ciliated epithelium
(B)	Inner surface of bronchiole	(II)	Areolar connective tissue
(C)	Blood	(III)	Cuboidal epithelium
(D)	Tubular parts of nephron	(IV)	specialised connective tissue

Choose the correct answer from the options

- A-III, B-IV, C-II, D-I
- A-I, B-II, C-IV, D-III
- A-II, B-III, C-I, D-IV
- A-II, B-I, C-IV, D-III

- Given below are two statements: (2023-Manipur)

Statement-I: Goblet cells are unicellular glands.

Statement-II: Earwax is the secretion of exocrine gland.

In the light of the above statements, choose the **correct** answer from the options given below:

- Statement-I** is true but **Statement-II** is false
- Statement-I** is false but **Statement-II** is true
- Both **Statement-I** and **Statement-II** are true
- Both **Statement-I** and **Statement-II** are false.

- Match **List-I** with **List-II**. (2023-Manipur)

List-I		List-II	
(A)	Columnar epithelium	(I)	Ducts of glands
(B)	Ciliated epithelium	(II)	Inner lining of stomach and intestine
(C)	Squamous epithelium	(III)	Inner lining of bronchioles
(D)	Cuboidal epithelium	(IV)	Endothelium

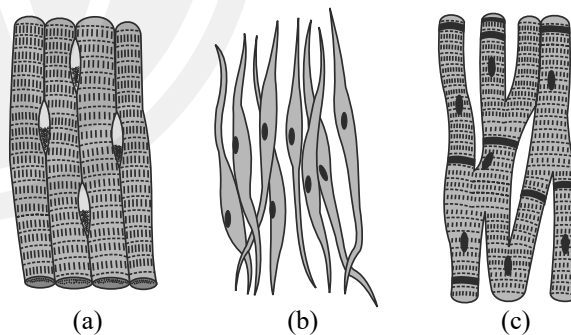
Choose the **correct** answer from the options given below:

- A-III, B-II, C-I, D-IV
- A-III, B-II, C-IV, D-I
- A-II, B-III, C-I, D-IV
- A-II, B-III, C-IV, D-I

9. Which of the following types of epithelium is present in the bronchioles and Fallopian tubes? (2022 Re)
 a. Stratified squamous epithelium
 b. Simple squamous epithelium
 c. Simple columnar epithelium
 d. Ciliated epithelium
10. Choose the correct statements: (2022 Re)
 A. Bones support and protect softer tissues and organs
 B. Weight bearing function is served by limb bones
 C. Ligament is the site of production of blood
 D. Adipose tissue is specialised to store fats.
 E. Tendons attach one bone to another.
 Choose the most appropriate answer from the options given below
 a. (A), (B) and (E) only b. (A), (B) and (D) only
 c. (B), (C) and (E) only d. (A), (C) and (D) only
11. Which of the following is present between the adjacent bones of the vertebral column? (2022)
 a. Smooth muscle b. Intercalated discs
 c. Cartilage d. Areolar tissue
12. Which of the following is not a connective tissue? (2022)
 a. Neuroglia b. Blood
 c. Adipose tissue d. Cartilage
13. Match List-I with List-II. (2022)
- | List-I | | List-II | |
|--------|----------------|---------|---------------------------------|
| (A) | Bronchioles | (i) | Dense Regular Connective Tissue |
| (B) | Goblet cell | (ii) | Loose Connective Tissue |
| (C) | Tendons | (iii) | Glandular Tissue |
| (D) | Adipose Tissue | (iv) | Ciliated Epithelium |
- Choose the correct answer from the options given below.
 a. A-iii B-iv C-ii D-i
 b. A-iv B-iii C-i D-ii
 c. A-i B-ii C-iii D-iv
 d. A-ii B-i C-iv D-iii
14. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules. (2021)
 a. Tight junctions and Gap junctions, respectively.
 b. Adhering junctions and Tight junctions, respectively.
 c. Adhering junctions and Gap junctions, respectively.
 d. Gap junctions and Adhering junctions, respectively.
15. Cuboidal epithelium with brush border of microvilli is found in: (2020)
 a. Ducts of salivary glands
 b. Proximal convoluted tubule of nephron
 c. Eustachian tube
 d. Lining of intestine
16. Goblet cells of alimentary canal are modified from: (2020)
 a. Columnar epithelial cells
 b. Chondrocytes
 c. Compound epithelial cells
 d. Squamous epithelial cells
17. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in (2019)
 a. Bile duct and bronchioles
 b. Fallopian tubes and pancreatic duct
 c. Eustachian tube and salivary duct
 d. Bronchioles and fallopian tubes
18. The function of the gap junction is to: (2015 Re)
 a. Facilitate communication between adjoining cells by connecting the cytoplasm for rapid transfer of ions, small molecules and some large molecules
 b. Separate two cells from each other
 c. Stop substance from leaking across a tissue
 d. Performing cementing to keep neighboring cells together.
19. Choose the correctly matched pair: (2014)
 a. Cartilage - Loose connective tissue
 b. Tendon - Specialised connective tissue
 c. Adipose tissue - Dense connective tissue
 d. Areolar tissue - Loose connective tissue
20. Choose the correctly matched pair: (2014)
 a. Inner surface of bronchioles - Squamous epithelium
 b. Inner lining of salivary ducts - Ciliated epithelium
 c. Moist surface of Buccal cavity - Glandular epithelium
 d. Tubular parts of nephrons - Cuboidal epithelium

Muscle Tissue & Neural Tissue

21. Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body: (2024)



Name of muscle/location

- a. (a) Skeletal - Biceps b. (a) Involuntary - Nose tip
 (b) Involuntary - Intestine (b) Skeletal - Bone
 (c) Smooth - Heart. (c) Cardiac - Heart.
 c. (a) Smooth - Toes d. (a) Skeletal-Triceps
 (b) Skeletal - Legs (b) Smooth - Stomach
 (c) Cardiac - Heart. (c) Cardiac - Heart.
22. Choose the correct statement about a muscular tissue: (2022 Re)
 a. Smooth muscles are multinucleated and involuntary.
 b. Skeletal muscle fibres are uninucleated and found in parallel bundles.
 c. Intercalated discs allow the cardiac muscle cells to contract as a unit.
 d. The walls of blood vessels are made up of columnar epithelium.

23. Which of the following statements **wrongly** represents the nature of smooth muscle? (2021)

- They are involuntary muscles
- Communication among the cells is performed by intercalated discs
- These muscles are present in the wall of blood vessels
- These muscle have no striations

24. Select the **incorrectly** matched pair from following: (2020-Covid)

- Neurons - Nerve cells
- Fibroblast - Areolar tissue
- Osteocytes - Bone cells
- Chondrocytes - Smooth muscle cells

25. Smooth muscles are: (2016 - II)

- Involuntary, cylindrical, striated
- Voluntary, spindle-shaped, uninucleate
- Involuntary, fusiform, non-striated
- Voluntary, multinucleate, cylindrical

26. Which type of tissue correctly matches with its location? (2016 - I)

	Tissue	Location
a.	Smooth muscle	Wall of intestine
b.	Areolar tissue	Tendons
c.	Transitional epithelium	Tip of nose
d.	Cuboidal epithelium	Lining of stomach

27. Which of the following is not a function of the skeletal system? (2015 Re)

- Storage of minerals
- Production of body heat
- Locomotion
- Production of erythrocytes

Cockroach

28. Which of the following is/are present in female Cockroach? (2024 Re)

- Collateral gland
- Mushroom gland
- Spermatheca
- Anal style
- Phallic gland

Choose the most appropriate answer from the options given below:

- B and D only
- B and E only
- A only
- A and C only

29. In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on: (2024)

- 8th and 9th segment
- 11th segment
- 5th segment
- 10th segment

30. Match List-I with List-II related to digestive system of cockroach. (2024)

List-I		List-II	
(A)	The structures used for storing of food.	(I)	Gizzard
(B)	Ring of 6-8 blind tubules at junction of foregut and midgut.	(II)	Gastric Caeca

(C)	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	(III)	Malpighian tubules
(D)	The structures used for grinding the food	(IV)	Crop

Choose the correct answer from the options given below:

- A-IV, B-III, C-II, D-I
- A-III, B-II, C-IV, D-I
- A-IV, B-II, C-III, D-I
- A-I, B-II, C-III, D-IV

31. Which of the following is characteristic feature of cockroach regarding sexual dimorphism? (2023)

- Presence of anal cerci
- Dark brown body colour and anal cerci
- Presence of anal styles
- Presence of sclerites

32. In cockroach, excretion is brought about by- (2023)

- Phallic gland
- Ureose gland
- Nephrocytes
- Fat body
- Collateral glands

Choose the correct answer from the options given below:

- B and D only
- A and E only
- A, B and E only
- B, C and D only

33. The Cockroach is: (2023-Manipur)

- Ammonotelic only
- Uricotelic only
- Ureotelic only
- Ureotelic and Uricotelic

34. Given below are two statements: (2023-Manipur)

Statement-I: In cockroach, the forewings are transparent and prothoracic in origin.

Statement-II: In cockroach, the hind wings are opaque, leathery and mesothoracic in origin.

In the light of the above statements, choose the **correct** answer from the options given below:

- Statement-I** is correct but **Statement-II** is incorrect
- Statement-I** is incorrect but **Statement-II** is correct
- Both **Statement-I** and **Statement-II** are correct
- Both **Statement-I** and **Statement-II** are incorrect

35. Match List-I with List-II regarding the organs of Cockroach: (2022 Re)

List-I		List-II	
(A)	Crop	(I)	grinding the food particles
(B)	Proventriculus	(II)	secretion of digestive juice
(C)	Hepatic caeca	(III)	removal of nitrogenous waste
(D)	Malpighian tubules	(IV)	storage of food

Choose the correct answer from the options given below.

- A-I, B-IV, C-III, D-II
- A-IV, B-I, C-II, D-III
- A-III, B-II, C-I, D-IV
- A-II, B-IV, C-I, D-III

36. Excretion in cockroach is performed by all, EXCEPT:

(2022 Re)

- Hepatic caeca
- Ureose glands
- Malpighian tubules
- Fat body

37. Tegmina in cockroach, arises from (2022)
 a. Prothorax and Mesothorax b. Prothorax
 c. Mesothorax d. Metathorax
38. Which of the following characteristics is **incorrect** with respect to cockroach? (2021)
 a. Hypopharynx lies within the cavity enclosed by the mouth parts.
 b. In females, 7th - 9th sterna together form a genital pouch.
 c. 10th abdominal segment in both sexes, bears a pair of anal cerci.
 d. A ring of gastric caeca is present at the junction of midgut and hind gut.
39. If the head of cockroach is removed, it may live for few days because: (2020)
 a. The cockroach does not have nervous system.
 b. The head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
 c. The head holds a 1/3rd of a nervous system while the rest is situated along the dorsal part of its body.
 d. The supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.
40. In cockroach, identify the parts of the foregut in correct sequence: (2020-Covid)
 a. Mouth → Crop → Pharynx → Oesophagus → Gizzard
 b. Mouth → Gizzard → Crop → Pharynx → Oesophagus
 c. Mouth → Pharynx → Oesophagus → Crop → Gizzard
 d. Mouth → Oesophagus → Pharynx → Crop → Gizzard
41. Match the following columns with reference to cockroach and select the correct option: (2020-Covid)

	Column-I		Column-II
1.	Grinding of the food particles	(i)	Hepatic caeca
2.	Secrete gastric juice	(ii)	10 th segment
3.	10 pairs	(iii)	Proventriculus
4.	Anal cerci	(iv)	Spiracles
		(v)	Alary muscles

- (1) (2) (3) (4) (1) (2) (3) (4)
 a. (iv) (iii) (v) (ii) c. (ii) (iii) (i) (iv)
 b. (i) (iv) (iii) (ii) d. (iii) (i) (iv) (ii)
42. Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth (2019)
 a. Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum
 b. Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum
 c. Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum
 d. Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum
43. Which of the following features is used to identify a male cockroach from a female cockroach? (2018)
 a. Presence of a boat shaped sternum on the 9th abdominal segment
 b. Presence of caudal styles

- c. Forewings with darker tegmina
 d. Presence of anal cerci
44. In male cockroaches, sperms are stored in which part of the reproductive system? (2016 - II)
 a. Testes b. Vas deferens
 c. Seminal vesicles d. Mushroom glands
45. Which of the following features is not present in *Periplaneta americana*? (2016 - I)
 a. Schizocoelom as body cavity
 b. Indeterminate and radial cleavage during embryonic development
 c. Exoskeleton composed of N-acetyl glucosamine
 d. Metamerically segmented body
46. The terga, sterna and pleura of cockroach body are joined by: (2015)
 a. Arthrodial membrane b. Cartilage
 c. Cementing glue d. Muscular tissue
47. The body cells in cockroach discharge their nitrogenous waste in the haemolymph mainly in the form of: (2015 Re)
 a. Potassium urate b. Urea
 c. Calcium carbonate d. Ammonia
48. Select the correct option with respect to cockroaches: (2013)
 a. Malpighian tubules convert nitrogenous wastes into urea
 b. Males bear short anal styles not present in females
 c. Nervous system comprises of a dorsal nerve cord and ten pairs of ganglia
 d. The forewings are tegmina which are used in flight
49. What external changes are visible after the last moult of a cockroach nymph? (2013)
 a. Labium develops
 b. Mandibles become harder
 c. Anal cerci develop
 d. Both fore wings and hind wings develop

Frog

50. Select the correct route for the passage of sperms in male frogs: (2017-Delhi)
 a. Testes → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca
 b. Testes → Vasa efferentia → Kidney → Seminal Vesicle → Urinogenital duct → Cloaca
 c. Testes → Vasa efferentia → Bidder's canal → Ureter → Cloaca
 d. Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca
51. Frog's heart when taken out of the body continues to beat for some time (2017-Delhi)
 Select the best option from the following statements
 A. Frog is a poikilotherm
 B. Frog does not have any coronary circulation
 C. Heart is "myogenic" in nature
 D. Heart is autoexcitable
 a. Only (A) b. Only (D)
 c. (A) & (B) d. (C) & (D)

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (d) | 4. (a) | 5. (c) | 6. (d) | 7. (c) | 8. (d) | 9. (d) | 10. (b) |
| 11. (c) | 12. (a) | 13. (b) | 14. (a) | 15. (b) | 16. (a) | 17. (d) | 18. (a) | 19. (d) | 20. (d) |
| 21. (d) | 22. (c) | 23. (b) | 24. (d) | 25. (c) | 26. (a) | 27. (b) | 28. (d) | 29. (d) | 30. (c) |
| 31. (c) | 32. (d) | 33. (b) | 34. (d) | 35. (b) | 36. (a) | 37. (c) | 38. (d) | 39. (b) | 40. (c) |
| 41. (d) | 42. (a) | 43. (b) | 44. (c) | 45. (b) | 46. (a) | 47. (a) | 48. (b) | 49. (d) | 50. (d) |
| 51. (d) | | | | | | | | | |

